Write Up Final Project Assembly

Eduardo Chavez Michel

Section 88090

MAE 214

Introduction

The goal of this project was to create and design a working shock absorber assembly system in SOLIDWORKS based on the reference material and the modeling skills we have learned throughout the semester. All key components for the shock absorber were modeled, materials were assigned, necessary dimensional adjustments were made, and a final assembly with an exploded view and Bill of Materials (BOM) was created.

Methods Used-CAD

Part Design

Every part-the bracket, spacer, U-Support, bushings, pivot, and spring- was created using the SOLIDWORKS software and advanced modeling techniques. Missing dimensions were properly determined by cross-referencing with other parts that provided insight into the measurements needed to make sure the assembly worked as intended with minimal to no error.

Material Selection

Materials were chosen based on research of common materials used within the context of automotive applications:

- <u>**Carbon Steel (Bracket, Spacer, U-Support, Pivot):**</u>

  These parts are the parts that carry the main structural elements of the assembly and must be able to withstand deformations, vibrations, and have high durability, while also being cost-effective
- <u>**Aluminum Bronze (Bushings):**</u>

  The bushings need to rotate in the assembly and must be able to handle the wear and tear that comes with normal conditions. Aluminum Bronze is commonly used for these applications as it withstands repeated motion.
- <u>**18-8 Stainless Steel (Shaft and Washer):**</u>

  The shaft must be corrosion-resistant and be able to resist wear. Stainless steel is a good choice for threaded parts that are subject to compressive forces and motion.
- <u>**Carbon Steel-Spring:**</u>

  The spring needs to be elastic and have the ability to deform while under load. Carbon steel is a good compression material.
- <u>**High Strength Steel (Hex Bolt, Lock Nut, Castle Nut, Cotter Pin):**</u>

  These fasteners are critical areas of our structure and require a high tensile strength. Making high-strength steel suitable.

Assembly Process:

All parts were assembled using appropriate mates: concentric, width, coincident, and skew. Dimensions were adjusted to prevent inferences that were possible. The video referenced served as a guide as to how the assembly should move and behave, and was used as a direct point of comparison to ensure correct assembly construction.

Engineering Drawings:

Each part has its corresponding part drawing, with the third-angle projection method being used and proper views being selected to ensure that there are no repetitive or repeated dimensions. The final assembly exploded view provides a detailed look into the parts and how they fit together to form the final product. As well as a bill of materials to account for a full, concise documentation of the assembly.

Results:

The final assembly behaves in accordance with the provided video. All of the sliding and rational structure elements fit and function correctly. A full set of engineering drawings was created for all parts of the assembly, and a final exploded view with a bill of materials was created. Fasteners were also properly selected and cataloged with their corresponding part numbers and measurements.

Concussion:

This project showcased the ability for us as engineers to model components, apply logical reasoning, materials reasoning, solve design gaps, and produce proper engineering drawings using the 3rd angle projection method. Completing this shock absorber assembly has made my skills in CAD become more diverse and well-rounded, developing my skillset further and allowing me to have better critical thinking within the space of computer-aided design. All in all, it has been a great semester, and I have learned a lot more about CAD than I had previously known. Thanks for making it a fun time.